Cross Validation on a Real Race

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```
# Setup
library(tidyverse)
library(Matrix)
library(rstan)

options(mc.cores = parallel::detectCores())
```

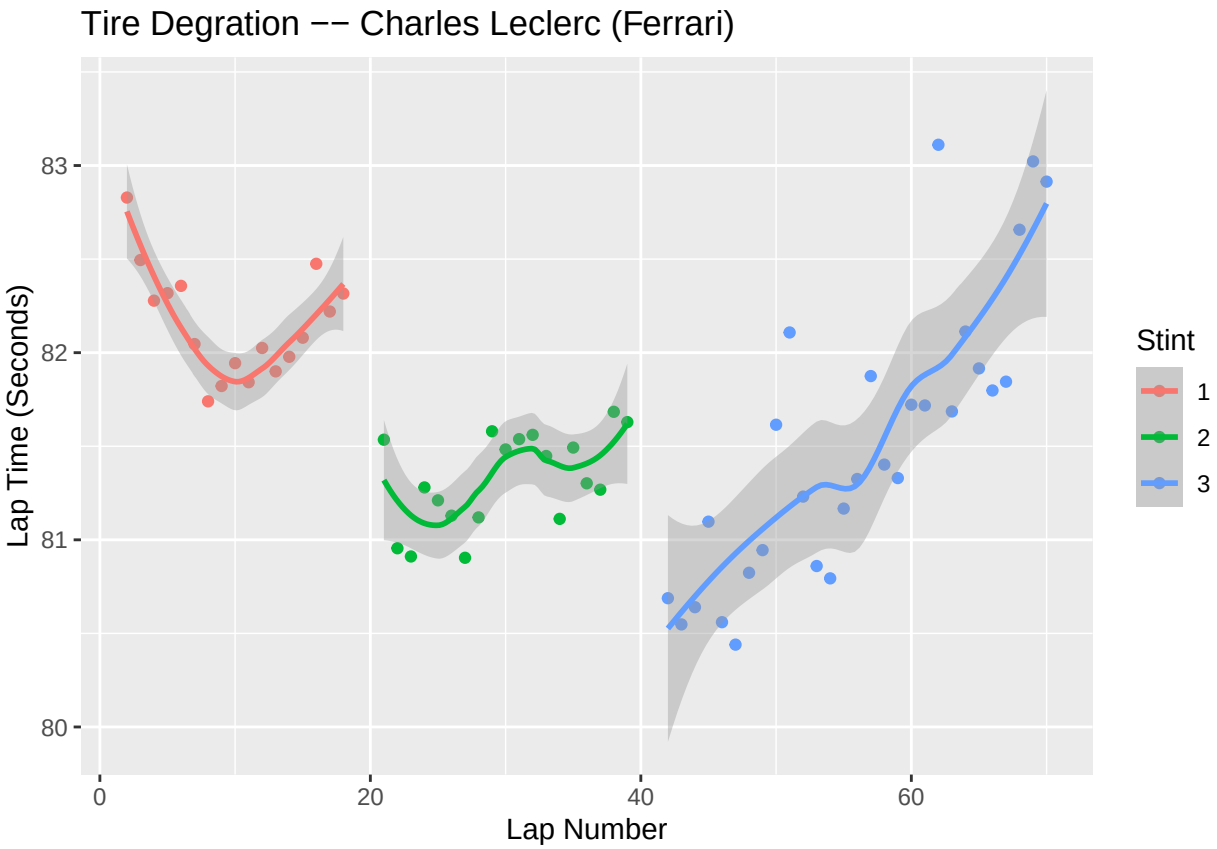
The Hugnarian Grand Prix 2025

I'm going to load lap time data from the Hungarian Grand Prix for various drivers. We'll do cross validation on Charles Leclerc's race because - while I admit I am slightly partial - he does seem to have a race that conforms relatively well to our model. Here is a visual of Leclerc's lap times:

```
hungary_laps <- read_csv("Hungary_laps.csv")

hungary_laps <- hungary_laps %>%
    mutate(Stint = as.factor(Stint),
           Compound = as.factor(Compound))

filter(hungary_laps, Driver == "LEC", is.na(PitOutTime) & is.na(PitInTime),LapTime < 100) %>%
    ggplot(mapping = aes(x = LapNumber, y = LapTime, colour = Stint)) +
    geom_point() +
    geom_smooth() +
    labs(title = "Tire Degration -- Charles Leclerc (Ferrari)",
         x = "Lap Number",
         y = "Lap Time (Seconds)")
```



Cross Validation Code

In this section I will include the code used for cross validation. Skip to the next section if interested in results.

Stan CV code

```
stint_CV <- function(model_string, data_list) {

    # Store the full data/time series
    y <- data_list[["y"]]
    # Store the full compound vector
    compound <- data_list[["Compound"]]
    # Store the full pit vector
    pit <- data_list[["Pit"]]
    # Store the Fuel vector
    fuel <- data_list[["fuel_mass"]]


    # Get the index of laps that end a stint
    end_stint <- which(data_list[["Pit"]] == 1)
    # Add the last lap because that also marks the end of a stint
    end_stint <- c(end_stint,length(y))

    # Number of Stints
    num_stints <- length(end_stint)
```

```r
# Will get RMSPE for each stint so need an empty vector
RMSPE_vec <- rep(NA, num_stints)

# Will get plots for each stint
plots <- vector("list", length = num_stints)

for(j in 1:num_stints) {

    # Do CV on last quarter of each stint
    # Number of folds
    if(j == 1) {
        K <- round(end_stint[j]/4)
    } else {
        K <- round((end_stint[j] - end_stint[j-1])/4)
    }

    stan.models <- vector("list", length = K)
    posteriors <- vector("list", length = K)

    # Vector to Store one-step ahead predictions
    ypred_hat <- rep(NA, K)

    for(i in 1:K) {

        # Update data vector to allow for next step prediction
        data_list[["TT"]] <- length(y[1:(i+end_stint[j]-K-1)])
        data_list[["y"]] <- y[1:(i+end_stint[j]-K-1)]
        data_list[["Compound"]] <- compound[1:(i+end_stint[j]-K-1)]
        c_map <- sort(unique(compound[1:(i+end_stint[j]-K-1)]))
        data_list[["compound_map"]] <- array(c_map, dim = length(c_map))
        data_list[["C_used"]] <- length(c_map)
        data_list[["Pit"]] <- pit[1:(i+end_stint[j]-K-1)]
        data_list[["fuel_mass"]] <- fuel[1:(i+end_stint[j]-K-1)]

        # Fit model on subset of data then get one step ahead prediction
        stan.models[[i]] <- stan(file = model_string,
                                 data = data_list,
                                 chains = 3, iter = 30000,
                                 control = list(adapt_delta = .99, max_treedepth = 12))
        posteriors[[i]] <- extract(stan.models[[i]])
        ypred_hat[i]  <- mean(posteriors[[i]]$y_pred)

    }

    # Plots
    df <- tibble(laps = 1:end_stint[j],
                 y = y[1:end_stint[j]])

    pred_df <- tibble(time = (end_stint[j] - K + 1):end_stint[j],
                      pred = ypred_hat)

    plots[[j]] <- ggplot() +
        geom_point(data = df,
```

```
                    aes(x = laps, y = y, color = "Observations")) +
          geom_point(data = pred_df,
                     aes(x = time, y = pred, color = "Predictions")) +
          scale_color_manual(values = c("Observations" = "red",
                                        "Predictions" = "blue")) +
          labs(title = paste("Scatter Plot of Lap vs Lap Times with Predictions for Stint", j),
               x = "Lap",
               y = "Laptime in Seconds",
               color = "Legend")

        # Calculate Summary Statistic
        MSPE <- mean((y[(end_stint[j]-K+1):end_stint[j]] - ypred_hat)^2)
        RMSPE_vec[j] <- sqrt(MSPE)

    }
    return(list(RMSPE_vec,plots))
}
```

Arima Baseline CV code

```
arima_cv <- function(y, pit) {

    # Get the index of laps that end a stint
    end_stint <- which(pit == 1)
    # Add the last lap because that also marks the end of a stint
    end_stint <- c(end_stint,length(y))

    # Number of Stints
    num_stints <- length(end_stint)

    # Will get RMSPE for each stint so need an empty vector
    RMSPE_vec <- rep(NA, num_stints)

    # Will get plots for each stint
    plots <- vector("list", length = num_stints)

    for(i in 1:num_stints) {

        # Number of folds
        if(i == 1) {
            K <- round(end_stint[i]/4)
        } else {
            K <- round((end_stint[i] - end_stint[i-1])/4)
        }

        arma.models <- vector("list", length = K)

        # Vector to Store one-step ahead predictions
        ypred_hat <- rep(NA, K)

        for(j in 1:K) {

            arma.models[[j]] <- arima(y[1:(j+end_stint[i]-K-1)], order = c(2,1,2))
```

```
            ypred_hat[j] <- as.numeric(predict(arma.models[[j]],n.ahead = 1)$pred)

        }
                # Plots
        df <- tibble(laps = 1:end_stint[i],
                     y = y[1:end_stint[i]])

        pred_df <- tibble(time = (end_stint[i] - K + 1):end_stint[i],
                          pred = ypred_hat)

        plots[[i]] <- ggplot() +
            geom_point(data = df,
                       aes(x = laps, y = y, color = "Observations")) +
            geom_point(data = pred_df,
                       aes(x = time, y = pred, color = "Predictions")) +
            scale_color_manual(values = c("Observations" = "red",
                                          "Predictions" = "blue")) +
            labs(title = paste("Scatter Plot of Lap vs Lap Times with Predictions for Stint", i),
                 x = "Lap",
                 y = "Laptime in Seconds",
                 color = "Legend")

        # Calculate Summary Statistic
        MSPE <- mean((y[(end_stint[i]-K+1):end_stint[i]] - ypred_hat)^2)
        RMSPE_vec[i] <- sqrt(MSPE)
    }

    return(list(RMSPE_vec,plots))

}
```

Analysis

Get the data ready

```
leclerc_df <- hungary_laps %>%
    filter(Driver == "LEC", is.na(PitOutTime) & is.na(PitInTime), LapTime < 100) %>%
    mutate(Compound_Code = as.integer(factor(Compound,levels = c("HARD","MEDIUM","SOFT"))))

leclerc_laps <- leclerc_df$LapTime

leclerc_compound <- leclerc_df$Compound_Code

leclerc_pit <- rep(0,length(leclerc_compound))

for(i in 2:length(leclerc_df$Stint)) {
    if(leclerc_df$Stint[i] != leclerc_df$Stint[i-1]) {
        leclerc_pit[i-1] <- 1
    }
}

fuel.kg <- seq(110,1,length.out = length(leclerc_laps))
```

```
leclerc_map <- sort(unique(leclerc_compound))
leclerc_used <- length(leclerc_map)
```

Cross Validation Results

```
# Arima cv
arima_results <- arima_cv(leclerc_laps, leclerc_pit)

# Stan SSM CV
leclerc_data <- list(TT = length(leclerc_laps), y = leclerc_laps, C = 3, Compound = leclerc_compound, C
stan_results <- stint_CV("full_race_1driver_and_fuel_real.stan", leclerc_data)
```

```
# Stint 1
arima_results[[1]][[1]]
```

```
## [1] 0.2807924
```

```
stan_results[[1]][[1]]
```

```
## [1] 0.2327326
```

```
# Stint 2
arima_results[[1]][[2]]
```

```
## [1] 0.3925258
```

```
stan_results[[1]][[2]]
```

```
## [1] 0.2163307
```

```
#Stint 3
arima_results[[1]][[3]]
```

```
## [1] 0.4572783
```

```
stan_results[[1]][[3]]
```

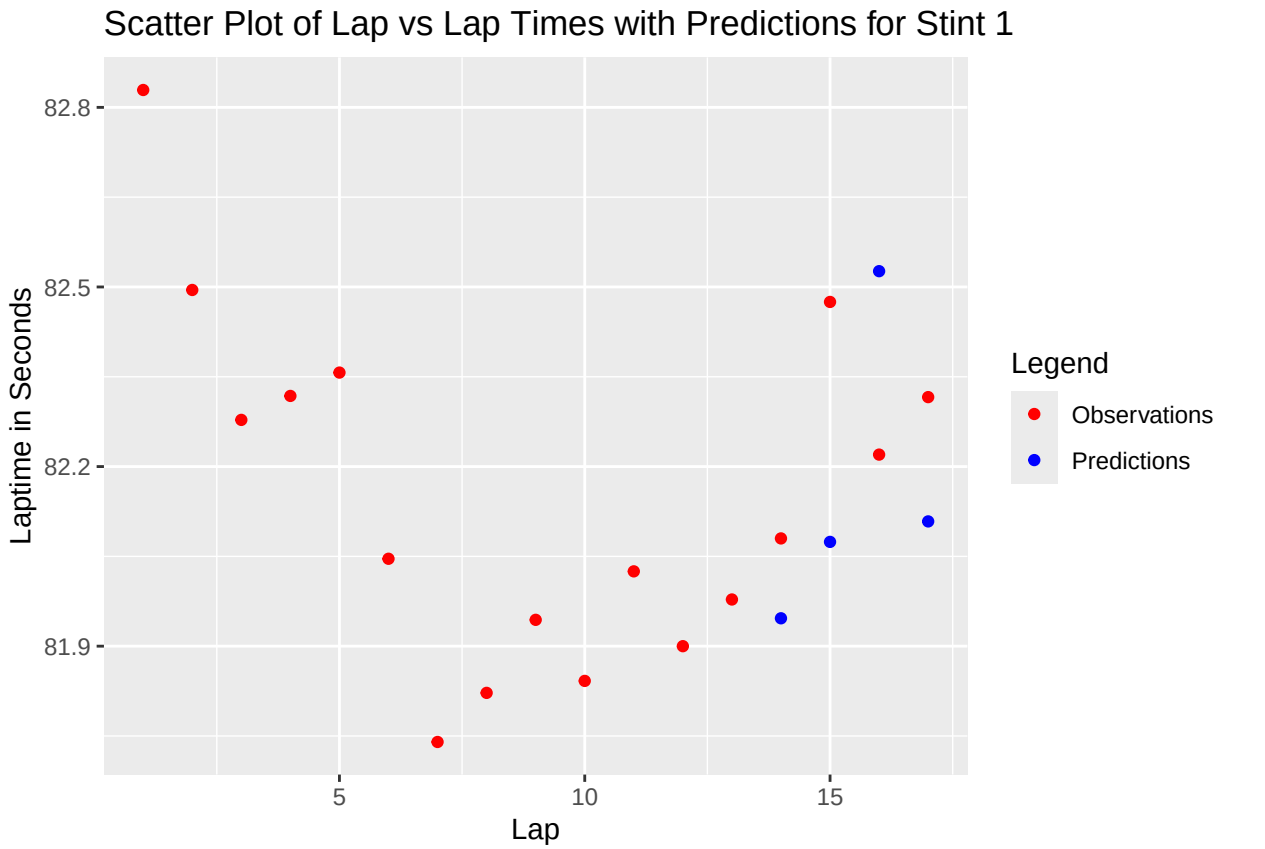
```
## [1] 0.4051673
```

Appendix

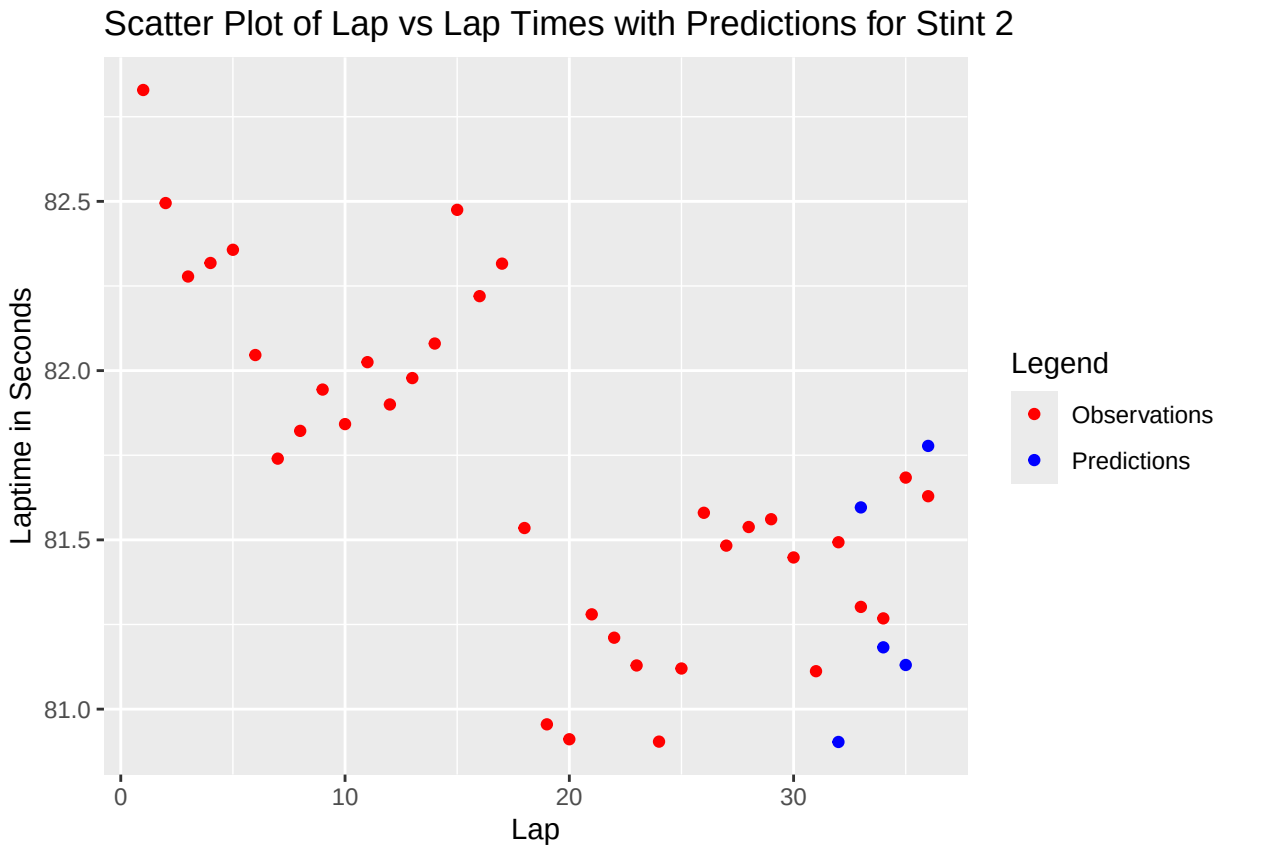
Plots of lap time vs lap with observations and predictions from both the arima model and our full SSM.

Arima Predictions

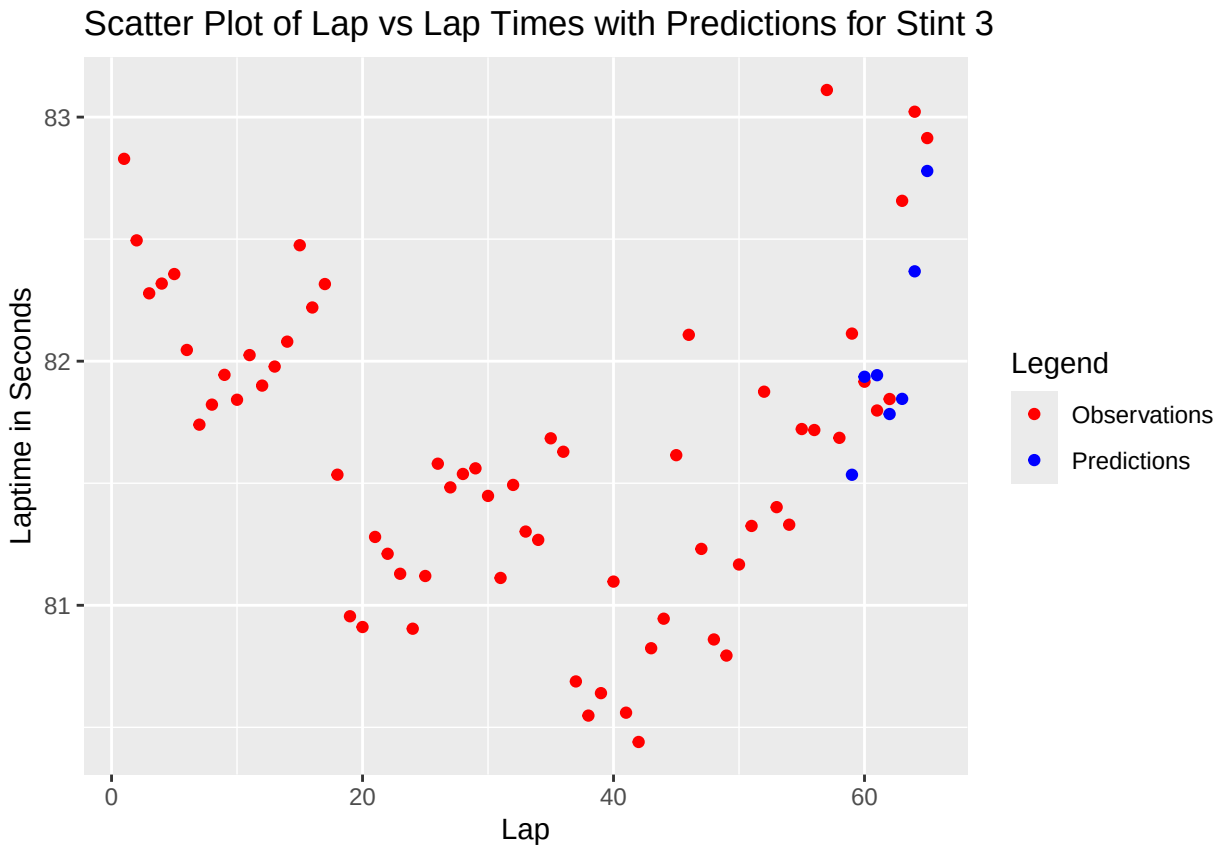
```
arima_results[[2]][[1]]
```




```
arima_results[[2]][[2]]
```

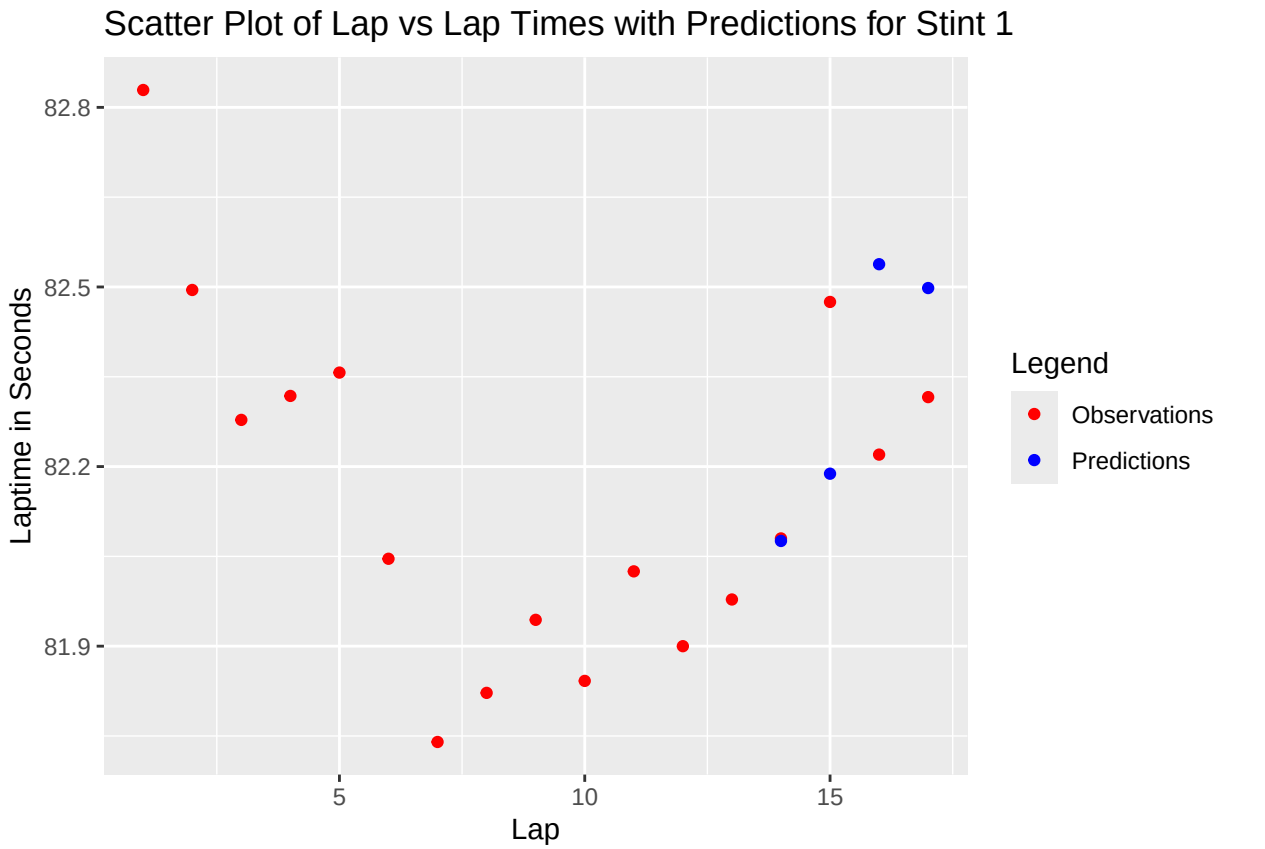



```
arima_results[[2]][[3]]
```

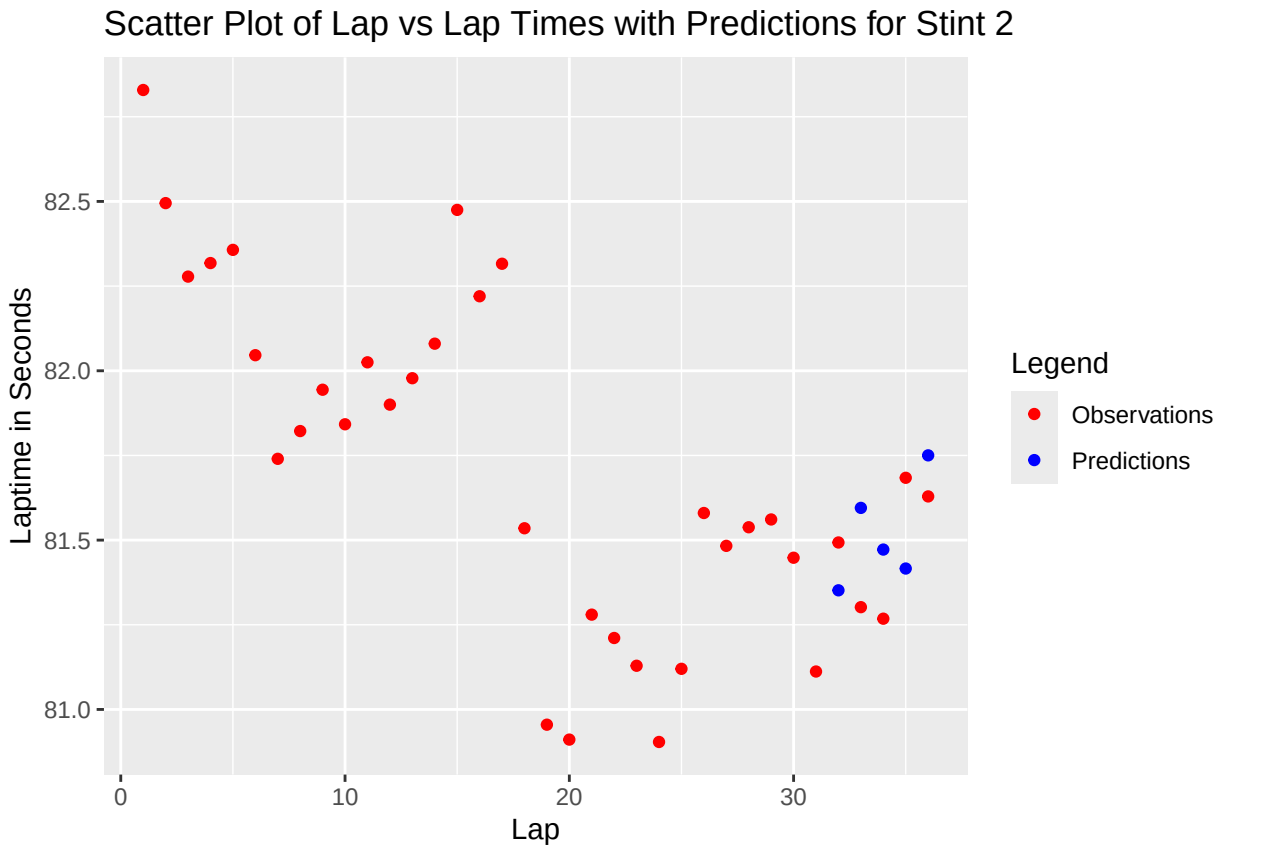


SSM Predictions

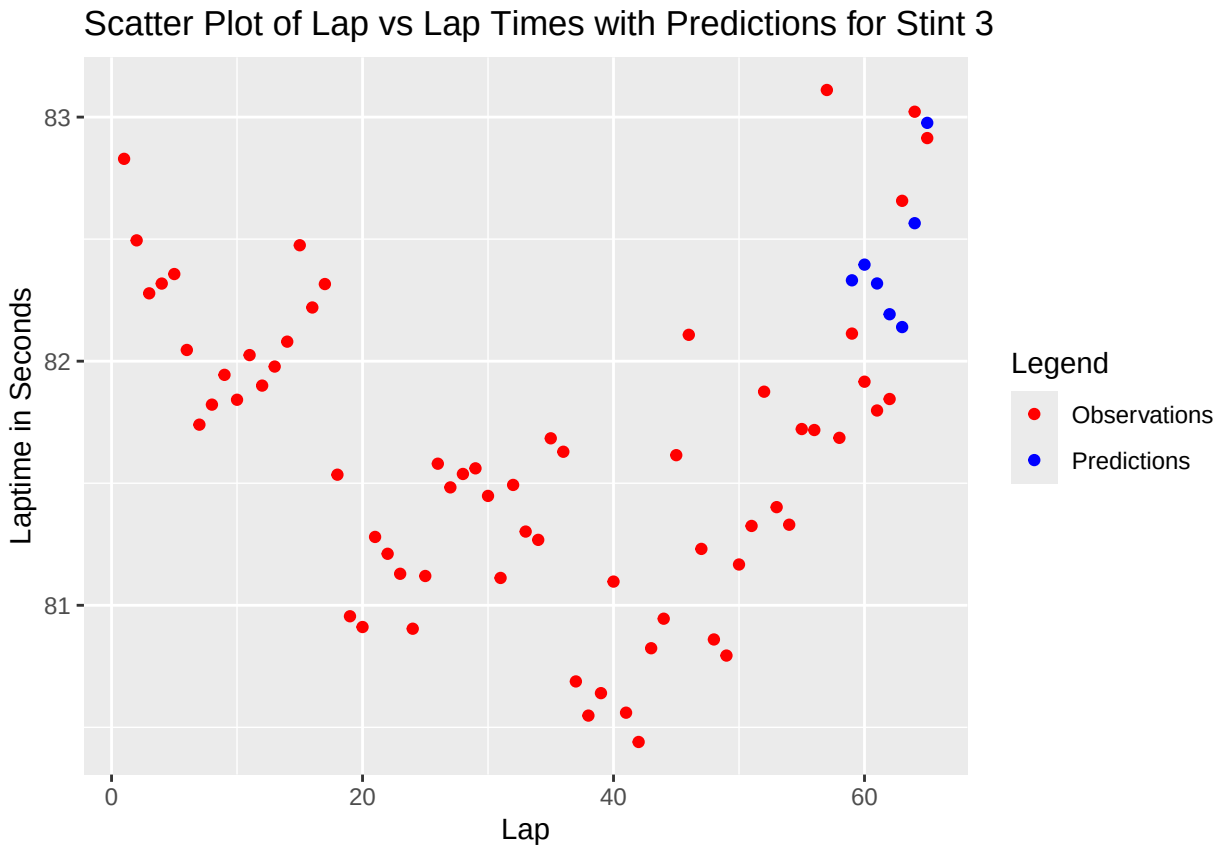
```
stan_results[[2]][[1]]
```




```
stan_results[[2]][[2]]
```




```
stan_results[[2]][[3]]
```


Scatter Plot of Lap vs Lap Times with Predictions for Stint 3
Laptime in Seconds
83
82
81
0
20
40
60
Lap
Legend
Observations
Predictions